Autostrux AI

Submission Features

- 🧠 Heuristic UR Optimization
- 📐 Tapered Sections Generation
- ✅ L/250 & H/150 Serviceability
- 🏗️ Full 3D Geometry & Accessories
- 📦 Automated BOQ & Costing

Developed for the Kaggle AI-Powered STAAD.Pro Hackathon.

🏗️ Autostrux AI: End-to-End PEB Generator

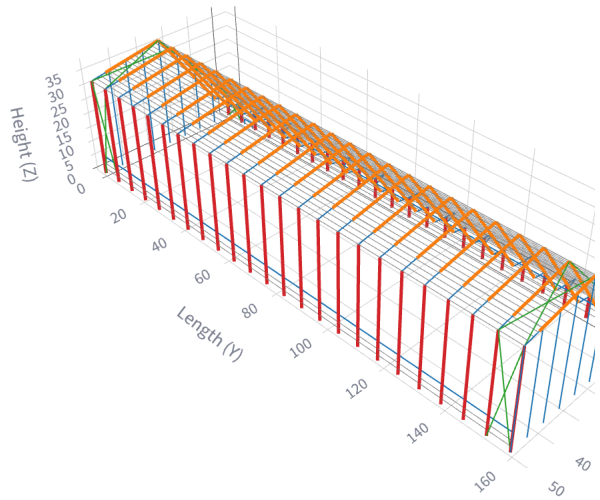
Automated 3D steel structure generator with analytical tapered section optimization.

📁 Upload Quote Request Form (JSON)

BulkStore.json
23.1KB

✅ File parsed successfully! Extracting geometry and load parameters...

🧊 Interactive 3D Geometry



⚙️ Analysis & Optimization

Start the Heuristic Engine to calculate Tapered sections, verify deflections, and generate the BOQ.

🚀 ESEGUI OTTIMIZZAZIONE E GENERA .STD

📊 Structural Results

Total Nodes	3D Members
1,122	1,147
Total Tonnage	**Max Utilization Ratio**
129.11 t	0.830
	↑ Optimized (Pass)

✅ Serviceability Checks (IS 800:2007)

Vertical Deflection: PASSED (Limit: L/250)

Lateral Sway: PASSED (Limit: H/150)

📥 Export Files

📐 Download .STD | 📦 Download BOQ

📋 Bill of Quantities (Takeoff Preview)

	Member	Type	Section	Length_m	Total_kg
0	1	COLUMN	TAPERED 0.300 0.006 0.400 0.200 0.010 0.200 0.010	6	
1	2	COLUMN	TAPERED 0.300 0.006 0.400 0.200 0.010 0.200 0.010	6	
2	3	HAUNCH	TAPERED 0.300 0.006 0.600 0.250 0.010 0.250 0.010	5.025	
3	4	RAFTER	TAPERED 0.500 0.006 0.300 0.225 0.010 0.225 0.010	20.1	
4	5	RAFTER	TAPERED 0.500 0.006 0.300 0.225 0.010 0.225 0.010	20.1	
5	6	HAUNCH	TAPERED 0.300 0.006 0.600 0.250 0.010 0.250 0.010	5.025	
6	7	ENDWALL_COL	WB200	6.625	
7	8	ENDWALL_COL	WB200	7.25	